

The empirical recurrence for OEIS A236636, recovered from its tail

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Abstract

OEIS A236636 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 474$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = -2293760 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236636 is “Number of $(n+1) \times (2+1) 0..2$ arrays colored with the maximum plus the upper median plus the minimum of every 2×2 subblock.”. It has offset 1 and begins

474, 4176, 37018, 334180, 2988228, 26755684, 239823278, 2152052046, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:23:35, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 474$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 474$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 948$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	0	c_{19}	13860325485	c_{37}	7519913804499	c_{55}	−10294405094381
c_2	6	c_{20}	12479231196	c_{38}	−7136899807550	c_{56}	−17022391477278
c_3	178	c_{21}	33253398013	c_{39}	10304454695527	c_{57}	−8510398010735
c_4	2754	c_{22}	68425003490	c_{40}	6312562472094	c_{58}	445501395167
c_5	10664	c_{23}	−160765498452	c_{41}	−19982384003924	c_{59}	3142973852613
c_6	34880	c_{24}	−220835030399	c_{42}	−4273160507023	c_{60}	2892842037995
c_7	77303	c_{25}	115433734126	c_{43}	−7517037770719	c_{61}	2179885285346
c_8	−412471	c_{26}	−237230553330	c_{44}	−5908811962186	c_{62}	1150654049312
c_9	−2037953	c_{27}	321050907227	c_{45}	24946555178688	c_{63}	379817181756
c_{10}	−3967652	c_{28}	1320586115370	c_{46}	22635279884283	c_{64}	101571593432
c_{11}	−8773273	c_{29}	−1105159069833	c_{47}	14011298635914	c_{65}	33618814560
c_{12}	25759784	c_{30}	−548033785846	c_{48}	−5441877645978	c_{66}	2287086272
c_{13}	167924437	c_{31}	1414796778922	c_{49}	−35209647619813	c_{67}	−4659807584
c_{14}	164171733	c_{32}	−3208615849398	c_{50}	−29605476603021	c_{68}	−1461902016
c_{15}	47721448	c_{33}	1423637630075	c_{51}	−7690457487846	c_{69}	−163179520
c_{16}	−674293605	c_{34}	4503533944043	c_{52}	19813222464360	c_{70}	−67101184
c_{17}	−5540376393	c_{35}	−7456694879365	c_{53}	36686785588559	c_{71}	−19993600
c_{18}	−4390358990	c_{36}	2045727478541	c_{54}	16805116453140	c_{72}	−2293760

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A236636 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = 2293760$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -2293760 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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